

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative practice(s) description

Course code and title: CS8592 Object Oriented Analysis and Design

Name of the Instructor(s): Mr.D.Lakshmanan, AP(S.G)/CSE

Date & Time: 17.09.2019 & 10.55AM to 11.45AM

Date: 17.09.2019

Innovative Practice:Simulation

Topic- Simlation of Mapping Design to Code using Argo UML

Objectives:

- To make the students to simulate and design with the UML dynamic and implementation diagrams.
- To make the students to generate java code from the UML class.

Justification for choosing this topic:

ArgoUML is an open source domain-oriented design environment that provides cognitive support of object-oriented design. The current version (0.30) of ArgoUML implements all the diagram types of the UML 1.4. It is written in Java and runs on every computer which provides a Java platform version 5 or newer. It uses the open file formats XMI (XML Metadata Interchange format) (for model information) and (Precision Graphics Markup Language) (for graph information) for storage. There are different types of association relationships are used in class diagrams. This tool generate the code automatically when the user draw the class diagram with different association. The simulation using Argo UML gives better understanding about the mapping design to code.

Activity:

There are four types of associations are explained by using Argo UML tool to simulate mapping design to code in the class room.

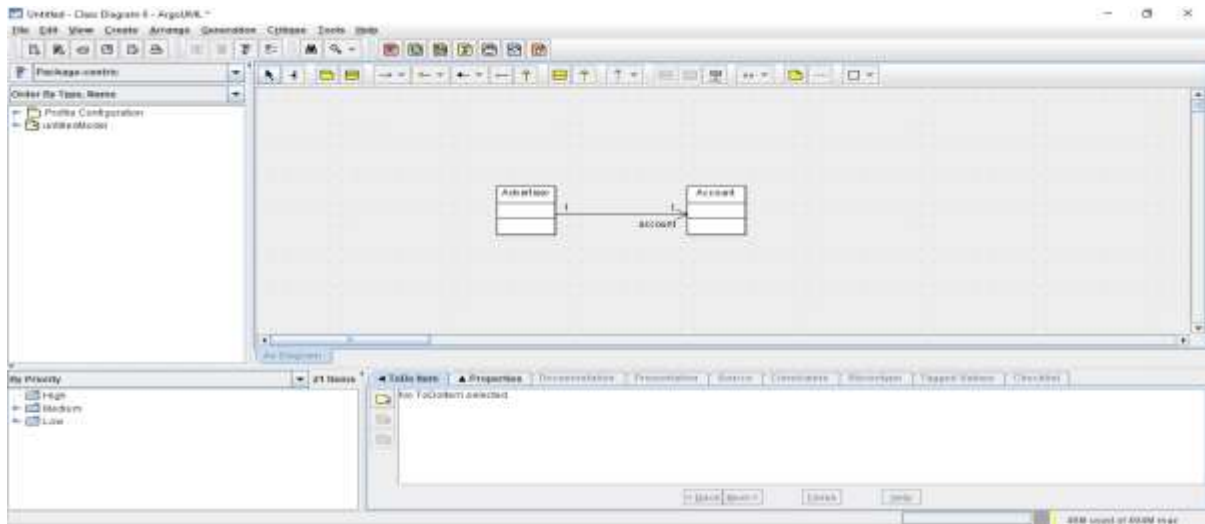
Mapping Associations

1. Unidirectional, one-to-one association
2. Bidirectional one-to-one association
3. Bidirectional, one-to-many association
4. Bidirectional, many-to-many association

Instructions to perform a simulation experiment.

- Open ArgoUML Tool
- Home page contains different menu such as File,Edit,View,Create, Arrange, usecase diagram icon,class diagram icon, sequence diagram,activity diagram and so on.
- Select class diagram
- In that window screen various symbols available to draw the class diagram are available
- On that select class symbol and different association symbols mentioned in the below diagrams.

1. Unidirectional, one-to-one association



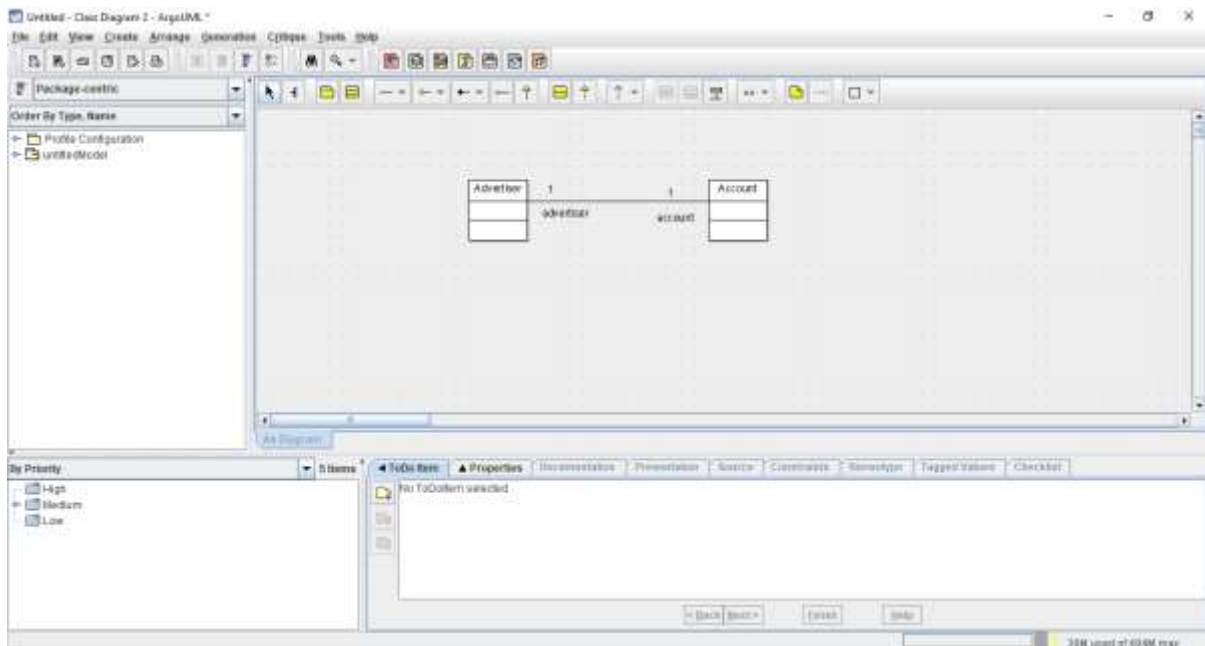
- After complete the diagram, To Generate all classes press F7 key in the Argo UML Tool.
- It will ask the language which we want to generate the code.
- Choose the java language.
- Choose the folder where we want to store the java file.
- After that it generate the java code as follows.

Generated Java code

```
public class Advertiser {  
    public Account account;  
}
```

```
public class Account {  
  
}
```

2. Bidirectional, one-to-one association



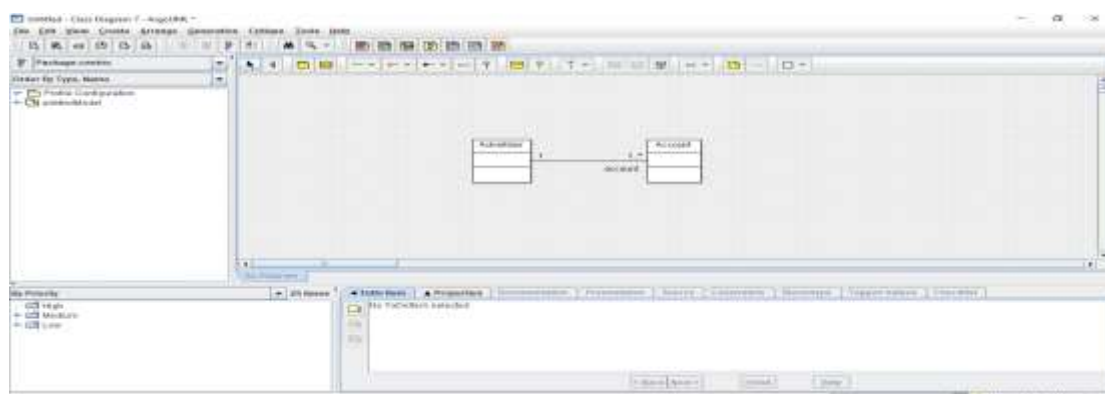
- After completing the diagram, to generate all classes press the F7 key in the Argo UML Tool.
- It will ask the language which we want to generate the code.
- Choose the Java language.
- Choose the folder where we want to store the Java file.
- After that it generates the Java code as follows.

Generated Java code

```
public class Account {  
    public Advertiser advertiser;  
}
```

```
public class Advertiser {  
    public Account account;  
}
```

3. Bidirectional, one-to-many association

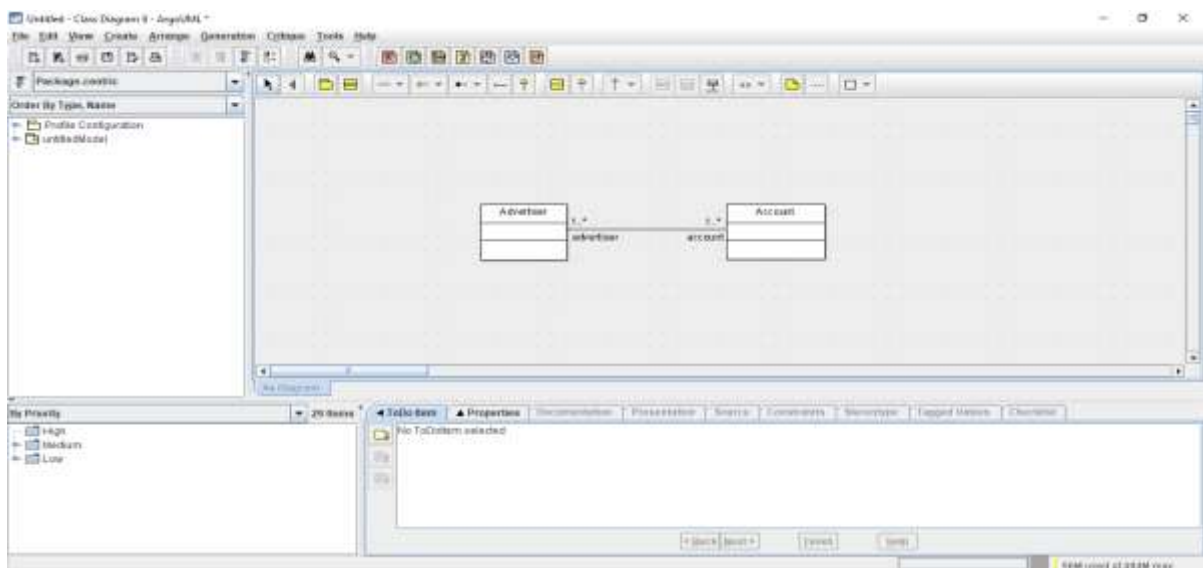


- After complete the diagram, To Generate all classes press F7 key in the Argo UML Tool.
- It will ask the language which we want to generate the code.
- Choose the java language.
- Choose the folder where we want to store the java file.
- After that it generate the java code as follows.

Generated Java code

```
import java.util.List;
public class Advertiser {
    public List<Account> account;
}
public class Account {
    public Advertiser myAdvertiser;
}
```

4. Bidirectional, many-to-many association



- After complete the diagram, To Generate all classes press F7 key in the Argo UML Tool.
- It will ask the language which we want to generate the code.
- Choose the java language.
- Choose the folder where we want to store the java file.
- After that it generate the java code as follows.

Generated Java code

```
import java.util.List;
public class Advertiser {
    public List<Account> account;
}
import java.util.List;
public class Account {
    public List<Advertiser> advertiser;
}
```



This activity helps in attaining the following CO – PO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO3	PO5
Transform UML based software design into pattern based design using design patterns.	2	2	2	3

SESSION OUTCOMES:

- This makes the students to generate the java class from the UML class diagram.
- It provides opportunities for the students to acquire knowledge about different types of association used to create java class.

Reference:

<http://argouml.tigris.org/>

Reflection Critique

Challenges:

Slow learners were not able to create java code from the class diagram. They felt very difficult to convert class diagram into code when association symbols are present in the class diagram.

Steps taken for address the challenges:

I used Argo UML tool to draw the class diagrams with different types of associations such as Unidirectional one-to-one association, Bidirectional one-to-one association, Bidirectional one-to-many association and Bidirectional many-to-many association.

Argo UML tool automatically generate the code for the class diagram in different languages such as JAVA, PHP and C# so on. Due to that students easily understand how to generate code from the class diagram by using this activity.

Prepared by:

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